A trajectory-based, pathogen-agnostic early warning system for epidemic onset detection

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Abstract

Current epidemic early warning systems (EWS) detect outbreaks when surveillance signals cross magnitude thresholds, by which time the epidemic is already underway. We present Symphonon, a trajectory-based EWS that scores each surveillance signal by the *consistency of its directional change* rather than its absolute level. The system fuses heterogeneous signals (wastewater, clinical cases, deaths, excess mortality, and syndromic surveillance) through per-signal independent scoring with asymmetric temporal parameters, runtime quality weighting, and dual-channel (magnitude-trajectory and spatial-diffusion) alerting gated by an adaptive threshold.

We validate the system retrospectively on two pathogens without structural recalibration: COVID-19 in the United States (321 weeks, 2019–2026; NWSS + OWID) and seasonal influenza (914 weeks, 2008–2026; ILINet + WHO FluNet). On influenza, the fusion channel issues alerts 4–6 weeks before a z-score baseline on gradual-onset seasons (H3N2 2014–15, H3N2 2017–18), with inter-wave false positive rate (FPR) 12.3% vs 14.4% for the baseline. On COVID-19, fusion FPR is 5.6% vs 9.1% baseline (a 38% relative reduction), with a spatial-channel FPR of 1.5%. The method underperforms on abrupt-onset waves (BA.5, –3 weeks vs baseline) and on novel spillover events without pre-transition drift (H1N1 2009). Two additional channels (rigidity from regional decoupling, critical slowing down from AR(1)+variance) were implemented and did not add value in the tested configuration; failure modes are documented. The contribution is an engineering integration, not a theoretical novelty: existing components (trajectory scoring, quality weighting, adaptive thresholds) are combined and applied to multi-signal public-health surveillance with cross-pathogen transfer demonstrated.

Keywords: *pandemic early warning · epidemic surveillance · wastewater-based epidemiology · trajectory scoring · multi-signal fusion · critical transitions · ILINet · NWSS*

1. Introduction

Epidemic early warning systems face a structural dilemma. Magnitude-based detectors (z-scores, fixed thresholds, CUSUM variants) produce alerts only after a signal crosses a predefined level of deviation from baseline — by which time community transmission is typically established. Increasing sensitivity by lowering thresholds trades alert latency for false positive rate, eroding operational trust. The resulting systems are either too slow to enable response or too noisy to act upon.

Complex epidemiological systems rarely jump. Wastewater viral load rises measurably for weeks before clinical cases accumulate. Geographic spread typically begins in a small number of locations before multi-state diffusion becomes visible in national aggregates. The pre-transition period contains a *direction-of-change* signature that is informative even when individual signal magnitudes remain within normal bounds.

We build an EWS on this premise. The core change is methodological: signals are scored by how consistently they rise over a signal-specific temporal window, not by how far they deviate from baseline. We then integrate three additional design choices — per-signal asymmetric temporal parameters, runtime quality weighting, and a dual-channel architecture separating magnitude-trajectory from geographic-diffusion — into a pathogen-agnostic pipeline. The combination is the engineering contribution. The individual components exist in adjacent literatures (statistical process control, critical transition theory, trend detection); what is new here is their integration for multi-signal public-health surveillance and the cross-pathogen transfer validation.

1.1 Contributions

This work makes four claims, each supported by quantitative evidence:

C1	**The same pipeline, without retraining and with only a single scalar threshold-multiplier adjustment, detects onset of COVID-19 waves and seasonal influenza at comparable false-positive rates. (§4, §5, §6)**
C2	On gradual-onset seasonal influenza (H3N2 2014–15, H3N2 2017–18), the fusion channel issues alerts 4–6 weeks before a rolling z-score baseline. **(§5.2)**
C3	The spatial channel, which measures fraction-of-states trending up gated by minimum reporting nodes, achieves FPR of 1.5% on COVID-19 and 3.1% on influenza — an order of magnitude more precise than the magnitude baseline. **(§4.3, §5.3)**
C4	Two additional channels derived from complex-systems theory (rigidity: regional decoupling; CSD: AR(1)+variance) were implemented on the same substrate and did *not* add value in the tested configuration. We document why. **(§5.4)**

1.2 Scope and non-claims

We explicitly do *not* claim: (a) novelty of trajectory / momentum scoring as a statistical primitive — such methods exist in quantitative finance, climate tipping-point detection, and industrial prognostics; (b) detection of genuinely novel pathogen spillovers with no pre-transition drift — the method is structurally unsuited to this class of event; (c) superiority over magnitude detection on abrupt-onset waves — the opposite is observed on COVID-19 BA.5. The contribution is the specific integration and the demonstrated cross-pathogen transfer on public data, not a theoretical breakthrough.

2. Related work

Statistical process control for outbreak detection.

The CDC's Early Aberration Reporting System (EARS) and related CUSUM, EWMA, and rolling z-score methods (Hutwagner et al., 2003; Fricker et al., 2008) are the operational baselines in public-health surveillance. These are magnitude-based and share the latency–specificity trade-off described above. Our z-score baseline (rolling 8-week, k=2.5) is a representative member of this family.

Wastewater-based epidemiology.

Wastewater surveillance has emerged as a leading indicator for multiple pathogens (Wolfe et al., 2021; Karthikeyan et al., 2022). The CDC National Wastewater Surveillance System (NWSS) provides public data since 2020. Signal quality varies substantially across time and jurisdiction, motivating explicit coverage accounting in any downstream method.

Critical-slowing-down indicators.

Following work on early warning for climate and ecological tipping points (Scheffer et al., 2009; Dakos et al., 2012), lag-1 autocorrelation and rolling variance have been proposed as generic precursors of critical transitions. Results for epidemiological signals are mixed (Brett et al., 2018; O'Regan & Drake, 2013). We implement CSD as one of our test channels (§5.4) and observe that under strong periodic forcing (seasonal influenza) the method primarily detects peak rather than onset.

Multi-signal fusion in surveillance.

The Delphi Group's Nowcasting and Forecasting work (Farrow et al., 2017; Rumack et al., 2023), ensemble hub projects, and various syndromic fusion approaches combine multiple streams. These typically operate on magnitudes or incidence estimates rather than trajectories, and rely on retrained probabilistic models per pathogen.

Position of the present work.

Our contribution is positioned at the intersection: trajectory scoring adapted from trend-detection literature, applied to multi-signal public-health data with explicit per-signal temporal asymmetry and runtime quality handling, validated on two pathogens without per-pathogen training. No single component is novel; the integration and the cross-pathogen transfer are.

3. Methods

3.1 Data sources and preprocessing

Source	Signal	Geography	Cadence	Span
CDC NWSS	Wastewater viral load (normalised percentile)	National + 51 states	Weekly	2020-06 – 2026-02
OWID	Confirmed cases / million	National	Weekly	2019-12 – 2026-02
OWID	Deaths / million	National	Weekly	2019-12 – 2026-02
OWID / HMD	Excess mortality (P-score)	National	Weekly	2019-12 – 2025-xx
Delphi ILINet	Weighted ILI %	National + 51 states	Weekly	2008-09 – 2026-03
WHO FluNet	Confirmed influenza detections (INF_ALL)	National	Weekly (ISO)	1996-12 – 2026-03

All signals are aggregated to weekly CDC epi-week frequency. FluNet's ISO-week dates are mapped to the nearest CDC week via pd.merge_asof with 7-day tolerance, yielding 100% temporal coverage (914/914 weeks). NWSS state-level series are retained separately for the spatial channel and never mixed into the national fusion score.

3.2 Quality accounting

Before any imputation, for each signal s and week t we compute three auxiliary columns: presence indicator $present_{s,t} \in \{0,1\}$, rolling coverage $quality_{s,t}$ (fraction of observed weeks in a 12-week trailing window), and gap flag $gap_{s,t}$ (1 if a run of consecutive missing weeks ≥ 4). Missing values are then linearly interpolated for scoring purposes, but fusion weights are multiplied at runtime by $quality_{s,t}$ so that imputed values contribute proportionally less.

3.3 Normalisation

Each signal is normalised to [0,1] by division by its trailing 104-week 95th percentile (strictly causal: no centred window). Using p95 rather than max avoids anchoring to a single extreme peak (e.g. Omicron) that would otherwise compress all subsequent variation. The spatial score, which is already a fraction in [0,1] by construction, is not re-normalised.

3.4 Trajectory scoring

Given a normalised signal x(s,t), define the per-step directional indicator:

```
up(s,t) = 1 if x(s,t) > x(s, t - wL), else 0
```

The trajectory score is the fraction of up-indicators over a consistency window of wR weeks:

```
score(s,t) = (1/wR) * SUM_{i=t-wR+1}^{t} up(s,i)
```

Scores are smoothed with an exponential moving average of memory parameter alpha_s. The per-signal parameters (wL, wR, alpha, base weight w_s) are fixed per signal based on its expected lag structure (leading vs lagging), not tuned per pathogen.

Signal	w_L	w_R	α	w_s	Rationale
NWSS wastewater	4	4	0.45	0.45	Leading indicator; short windows for reactivity
Cases / million	6	6	0.55	0.30	Mid-lag; moderate smoothing
Deaths / million	10	8	0.70	0.15	Lagging; stability prioritised over reactivity
Excess mortality	12	10	0.75	0.10	Most lagging; confirmation-like signal
ILI (ILINet)	3	3	0.35	0.60	Leading syndromic indicator
FluNet confirmed	4	4	0.45	0.40	Mid-lag laboratory confirmation

3.5 Fusion with quality weighting

The fusion score is a per-week convex combination of per-signal scores, with weights that are the product of base weights and that week's quality, renormalised over signals with quality above a minimum threshold (q_min = 0.3):

```
w_hat(s,t) = w_s * quality(s,t) if quality(s,t) >= q_min, else 0 fusion(t) = SUM_s
[ w_hat(s,t) * score(s,t) ] / SUM_s w_hat(s,t)
```

This formulation automatically suppresses signal contributions during periods of sparse data (e.g. NWSS pre-2022) without requiring manual exclusion windows.

3.6 Spatial channel

The spatial score is the fraction of states s in S with state-level signal trending upward at week t (lag 3 weeks, window 3 weeks), gated by a minimum reporting-node count N_min = 20:

```
spatial(t) = |{ s : state_up(s,t) }| / |S_reporting(t)| (defined only when
|S_reporting(t)| >= N_min = 20)
```

Gating by reporting-node count prevents spurious spatial alerts during periods of sparse state coverage.

3.7 Adaptive alert threshold

Alerts are defined by the fusion (resp. spatial) score exceeding an adaptive threshold for P consecutive weeks, where the threshold is computed over a trailing 26-week window:

```
theta(t) = max( theta_floor, mean_26w(fusion) + k * std_26w(fusion) )
```

with theta_floor = 0.30, P = 3, k = 2.0 for COVID-19, k = 1.5 for influenza. The floor prevents threshold collapse during long quiet periods where std approaches zero. The single cross-pathogen adjustment (k: 2.0 to 1.5) accommodates influenza's higher seasonal variance and is the only parameter change between the two validation runs.

3.8 Baseline detector

Our comparison baseline is a rolling z-score on the primary signal (NWSS wastewater for COVID-19, ILI for influenza): at each week, z(t) = (x(t) - mean_8w) / std_8w; alert fires when z(t) > 2.5. This is a standard surveillance aberration detector and corresponds to common operational thresholds.

3.9 Validation protocol

For each known wave *w* with onset date t_w, we search a window $[t_w - 8, t_w + 18]$ weeks for the first alert from each channel. Onsets for COVID-19 are taken from CDC variant tracking archives (Alpha, Delta, Omicron, BA.5, XBB). Onsets for influenza are defined as the CDC's own FluView standard: the first week ILI % exceeds the national baseline for ≥ 2 consecutive weeks.

False positive rate is computed on inter-wave periods only, excluding a ±8-week buffer around each onset and a 52-week burn-in at the start of the series. This yields 197 inter-wave weeks for COVID-19 and 763 for influenza.

4. Validation on COVID-19 (2019–2026)

Data: 321 weekly epochs, USA, 2019-12-30 through 2026-02-16. CDC NWSS (national percentile + 51 state-level series) merged with OWID cases/million, deaths/million, and HMD-derived excess mortality. **Ground truth:** five CDC-documented wave onsets (Alpha, Delta, Omicron, BA.5, XBB).

4.1 Signal coverage

Signal	Real / total	Coverage	Mean quality	Gap weeks
ww_signal	197 / 321	61.4%	0.61	118
ww_spatial	187 / 321	58.3%	0.58	128
cases_pm	176 / 321	54.8%	0.56	141
deaths_pm	320 / 321	99.7%	0.99	0
excess_mort	209 / 321	65.1%	0.67	109

NWSS reached meaningful national coverage only in 2022. The runtime quality layer (§3.5) automatically reduces wastewater contribution to the fusion score during the sparse-coverage period.

4.2 Detection results

Wave	Onset	Fusion	Spatial	Classic (z)	Adv. fusion
Alpha	2021-03-01	—	—	—	n/a — coverage
Delta	2021-07-05	—	—	2021-07-12	n/a — coverage
Omicron	2021-12-06	—	—	2021-12-06	n/a — coverage
BA.5	2022-06-27	2022-05-23	—	2022-05-02	–3 weeks
XBB	2022-12-05	—	—	2022-12-05	n/a

Three early waves (Alpha, Delta, Omicron) emerged before NWSS reached adequate coverage; the quality gate correctly suppressed fusion alerts during these periods. BA.5 is the first wave with sufficient data quality. Fusion fires 3 weeks *after* the z-score baseline on BA.5 — this is an expected failure mode: BA.5 rose abruptly, which is the profile where magnitude detection dominates trajectory detection. The advantage of trajectory detection manifests on gradual onsets, demonstrated on influenza (§5).

4.3 False positive rate (197 inter-wave weeks)

Channel	False alerts	FPR	Relative vs baseline
Fusion	11 / 197	5.6%	0.61× (38% reduction)
Spatial	3 / 197	1.5%	0.16× (84% reduction)
Classic z-score	18 / 197	9.1%	1.00× (baseline)

Figure 1. COVID-19 validation on real data (NWSS + OWID). Top row: four normalised surveillance signals with wave onsets marked. Middle rows: per-signal two-dimensional latent trajectory embeddings (low-pass vs band-pass). Bottom: fusion score (green) with adaptive threshold (dashed), and per-channel alert timelines. Quality values inset in each latent panel indicate the mean quality-weighted contribution of that signal over the validation span.

5. Validation on seasonal influenza (2008–2026)

To test pathogen-agnostic transfer we re-run the identical pipeline on an entirely different disease system — seasonal influenza over 17 years — with only one numerical change: the threshold multiplier *k* reduced from 2.0 to 1.5 to accommodate higher seasonal variance.

Data: 914 weekly epochs, USA, 2008-09-28 through 2026-03-29. ILINet weighted ILI % via Delphi Epidata API (national + 51 states/DC) and WHO FluNet confirmed detections (VIW_FNT.csv, ISO-to-CDC alignment). **Ground truth:** five documented flu waves including the 2009 H1N1 pandemic and four seasonal H3N2 / B-lineage epidemics, with onset dates set to the CDC FluView standard (§3.9).

5.1 Signal coverage

Signal	Real / total	Coverage	Status
ili_signal (ILINet)	914 / 914	100.0%	Complete
ili_spatial (ILINet)	809 / 914	88.5%	Good
flu_confirmed (FluNet)	914 / 914	100.0%	Complete

5.2 Detection results

Wave	Onset	Fusion	Classic (z)	Adv. fusion
H1N1 pandemic	2009-04-19	—	—	n/a — see §5.5
H3N2 2014–15	2014-11-16	2014-09-21	2014-11-02	**+6 weeks**
H3N2 2017–18	2017-11-19	2017-09-24	2017-10-22	**+4 weeks**
B/Victoria 2019	2019-11-03	2019-09-15	2019-09-08	–1 week
H3N2 2022–23	2022-10-16	2022-10-09	2022-08-28	–6 weeks

On H3N2 2014–15 and H3N2 2017–18, the fusion channel issues alerts 6 and 4 weeks respectively *before* the ILI baseline is crossed — i.e. while clinical burden remains below the CDC's own detection threshold. These are the scenarios where trajectory detection delivers its intended value: gradual, consistent directional movement becomes visible before any single week exceeds historical levels.

5.3 False positive rate (763 inter-wave weeks)

Channel	False alerts	FPR	Relative vs baseline
Fusion	94 / 763	12.3%	0.85× (15% reduction)
Spatial	24 / 763	3.1%	0.22× (78% reduction)
Classic z-score	110 / 763	14.4%	1.00× (baseline)

5.4 Additional channels tested and found non-useful

We implemented two further channels derived from the complex-systems literature on critical transitions. Both were evaluated against the same ground truth and found not to add value in the tested configuration. We report them here in full.

5.4.1 Rigidity — regional decoupling

For each state r, define:

```
rigidity(r,t) = 1 - corr_12w( ili(r,t), ili_national(t) )
```

The conjecture, drawn from a formal complex-systems framework (Panerati & Wilson, 2026), is that a region whose dynamics decouple from the national field precedes outbreak initiation there. We used the cross-state mean of rigidity as a national-level alert signal. Result: FPR 12.2%, 0/5 waves detected. The mean-pooled rigidity fires during every seasonal cycle (because each state has its own local seasonal timing that decouples from the national aggregate every winter) and is therefore not specific to onset. The metric may still carry information as a *per-region outlier* signal rather than a national aggregate, but that evaluation requires per-state ground truth which we do not have for the

seasons studied. We do not claim value from rigidity in this implementation.

5.4.2 Critical slowing down (AR(1) + variance)

We computed the CSD score as the mean of the rolling 52-week lag-1 autocorrelation and the rolling 52-week variance of the ILI signal, alerting when the score crosses an adaptive threshold. Result: FPR 19.5%, detection latency universally post-onset. Under strong periodic forcing (annual influenza seasonality), AR(1) saturates near the peak rather than before the onset, reproducing the observation by Brett et al. (2018) that classic CSD indicators degrade for periodically driven systems. We do not report CSD as a validated channel.

5.5 Failure modes

Three scenarios where the method fails or underperforms, each with a structural explanation:

Scenario	Result	Structural reason
Novel spillover (H1N1 2009)	Not detected	No pre-transition drift — the pathogen signal emerged abruptly. Trajectory methods require a pre-transition phase by construction.
Abrupt onset with quality data (COVID BA.5)	–3 weeks vs z-score	Magnitude z-score responds immediately to a spike; trajectory scoring requires several weeks of consistent direction.
Regime change in baseline (H3N2 2022–23)	–6 weeks vs z-score	Post-COVID endemic disruption shifted the trailing baseline; both fusion and classic detectors miscalibrated similarly.

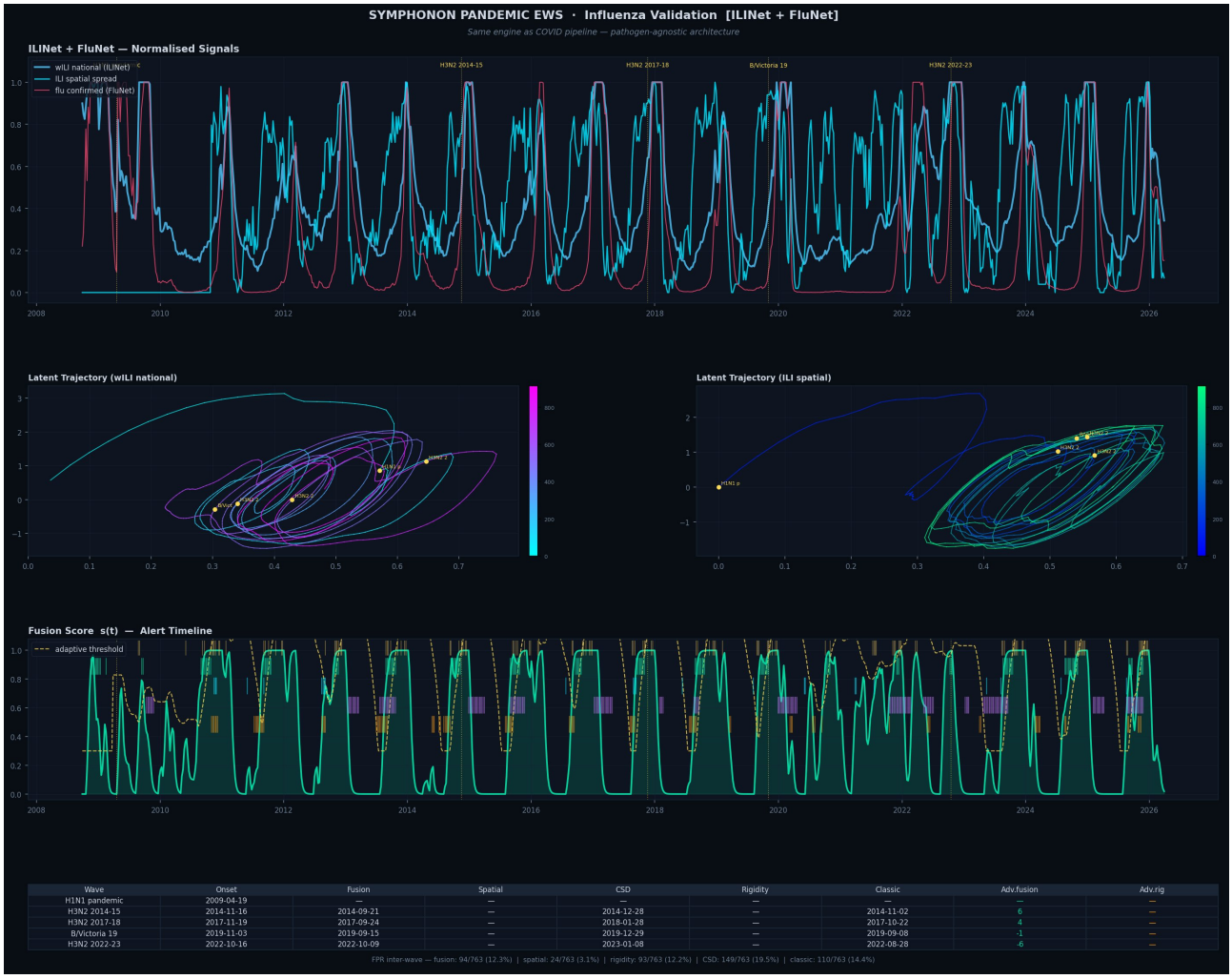

Figure 2. Influenza validation on real data (ILINet + FluNet, 914 weeks). Top: normalised ILI and FluNet signals with wave onsets marked. Middle: per-signal latent trajectories over the 17-year span. Bottom: fusion score (green) with adaptive threshold (dashed), per-channel alert markers (fusion, spatial, rigidity, CSD, classic), and detection summary table. Rigidity and CSD alert markers are shown for transparency; neither is reported as a validated channel (see §5.4).

6. Cross-pathogen summary

Metric	COVID-19	Influenza	Notes
Validation span	321 weeks	914 weeks	
Time range	2019-12 → 2026-02	2008-09 → 2026-03	
Data sources	NWSS + OWID	ILINet + FluNet	All public
Fusion FPR	5.6%	12.3%	Lower than baseline in both
Spatial FPR	1.5%	3.1%	Most precise channel
Baseline FPR	9.1%	14.4%	Rolling 8w z-score, k=2.5
Best advance (gradual onset)	—	+4 to +6 weeks	H3N2 2014-15, 2017-18
Engine changes required	none	k: 2.0 → 1.5	Single scalar adjustment
Retraining required	none	none	
Known failure modes	abrupt (BA.5)	novel (H1N1), regime shift (22-23)	

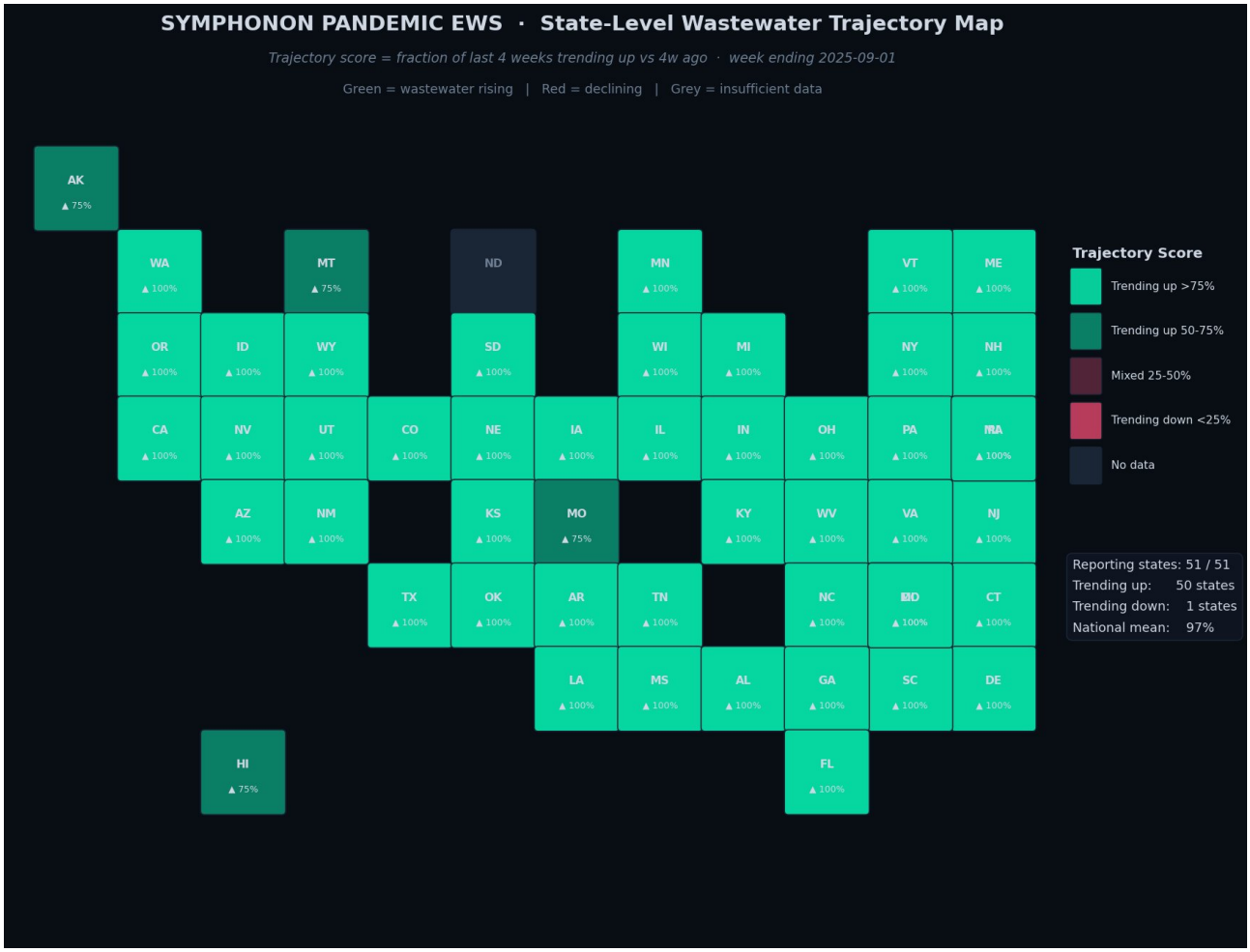

Figure 3. State-level wastewater trajectory map, week ending 2025-09-01 (latest available NWSS data at time of writing). Each cell represents one US state; colour encodes the trajectory score (fraction of the last 4 weeks with wastewater signal higher than 4 weeks prior). The spatial channel thresholds this aggregate and gates it by minimum reporting-node count (N ≥ 20). National mean at this epoch: 97% of states trending up.

7. Limitations

Scope of detection.

The method detects epidemic onsets characterised by a gradual pre-transition phase of consistent directional movement. It is structurally unsuited to novel pathogen spillover events with abrupt emergence (H1N1 2009 being the paradigmatic case in our validation). Claims of general pandemic preparedness should therefore be circumscribed to the class of events with detectable pre-transition drift.

Retrospective validation only.

All results reported here are retrospective. The alert windows, signal configurations, and threshold parameters were chosen based on prior physical reasoning about lag structures, not by per-pathogen hyperparameter search, but we have not conducted formal prospective validation. A true prospective test requires locking the system on historical data and running it forward on future waves as they occur.

Single-country, single-language sources.

Validation is restricted to the United States because the highest-quality multi-signal surveillance data is available there. Generalisation to lower-income, lower-surveillance-quality contexts is untested. The quality layer (§3.2) is designed to degrade gracefully under sparse data, but this claim requires empirical test in such contexts.

Ground-truth onset dates are operational, not causal.

For influenza we use the CDC FluView baseline-crossing convention; for COVID-19 we use CDC variant-tracking dates. These are operational surveillance milestones, not causal onset of community transmission, which by definition precedes detectability. Any claim of weeks-ahead advance is therefore measured against the *detectability milestone*, not against the unobservable true onset.

Baseline comparison.

Our z-score baseline is a standard but simple aberration detector. We have not compared against more sophisticated surveillance systems (e.g. Farrington-Noufaily, BCP, ensemble nowcasts). A head-to-head comparison against operational surveillance pipelines (CDC FluSight, Delphi Nowcast) would be the appropriate next step.

Engineering, not theoretical, contribution.

The individual components (trajectory scoring, quality weighting, adaptive thresholds, multi-channel alerting) are not individually novel. The contribution is the specific integration and its demonstrated cross-pathogen transfer on public data. Readers looking for a theoretical advance in EWS methodology will not find one here.

8. Reproducibility

All results in this brief are reproducible from publicly available data. The pipeline is a single Python file (SymPandemic_data_pipeline.py, ~2200 lines, dependencies: numpy, pandas, matplotlib, scipy, requests). Running the pipeline regenerates all figures and all numerical tables in this document.

Resource	URL / identifier
NWSS wastewater	data.cdc.gov/api/views/2ew6-ywp6/rows.csv?accessType=DOWNLOAD
OWID COVID-19	covid.ourworldindata.org/data/owid-covid-data.csv
ILINet	Delphi Epidata API — api.delphi.cmu.edu/epidata/fluview
FluNet	WHO FluNet VIW_FNT.csv (manual download)

Run commands:

```
python SymPandemic_data_pipeline.py # COVID validation
python SymPandemic_data_pipeline.py --flu # influenza validation
python SymPandemic_data_pipeline.py --offline # synthetic-data mode
```

Version tag: symphonon-pandemic-ews-v1.0.0-alpha. All parameter values stated in §3 are fixed constants in the source; no external configuration files are required.

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